CYP inhibition prediction through integrated machine learning and structural analysis

We combined machine learning (ML), Canopy property predictions, chemistry-aware descriptors and our bespoke CYP docking to predict direct-inhibition pIC50 for CYP1A2, CYP2C9, CYP2D6 and CYP3A4. The workflow provides predicted pIC50 values and inspectable binding poses for compound design. We used 6,525 endpoint measurements from 4,905 compounds and generated four-isoform predictions for 750 blinded compounds. Higher pIC50 indicates stronger inhibition.

ML

We used random forest (RF) models as a baseline, first with molecular fingerprints and then with added chemistry descriptors. We tested XGBoost (XGB) on the same inputs, but it did not consistently outperform RF.

We then trained Chemprop networks on molecular structures. Multitask models learned several CYP and assay responses together. They improved CYP2D6 development results over the single-task network, although adding more tasks did not always help. We averaged selected single-task and multitask predictions. In the final model, XGB corrects those predictions using chemistry, Canopy and docking features, with each correction limited to less than one pIC50 unit. A final CYP2D6 calibration lowers the prediction mean by 0.5 and sets its spread to 0.85 times the training-label standard deviation.

Canopy

We calculated properties and inhibition scores with Canopy, our in-house software at Molecular Forecaster Inc. (MFI). We supplied 41 Canopy values, including molecular weight, logP, logD7.4, logS, polar surface area, hydrogen-bond counts, ring properties, molecular charge and predicted biological properties. These summarize solubility, lipophilicity, polarity and molecular size, which affect protein-ligand recognition.

The inputs also include five CYP inhibitor scores. These scores use an IC50 threshold below 10 μM, equivalent to pIC50 above 5. We used them as continuous descriptors alongside the other properties. This gives the regression models additional information about inhibition likelihood while retaining their quantitative pIC50 outputs.

Chemistry-aware features

We added 64 descriptors covering amine classes, ionizable nitrogen, formal charge, acidic groups, aromatic and fused rings, heterocycles, molecular size and flexibility. These features make relevant chemical distinctions explicit. CYP2D6 commonly recognizes basic nitrogen-containing ligands; CYP2C9 often recognizes acidic groups; CYP1A2 accommodates planar aromatic structures; CYP3A4 accepts a broader range of sizes and shapes.

We also encoded distances between functional groups and aromatic systems, along with combinations such as aromatic content and lipophilicity. These descriptors help the models distinguish compounds with similar size or composition but different functional-group placement.

Docking

We used **our bespoke CYP docking workflow in FITTED, including the iron-coordination scoring described by Labarre et al., 2022**. Its Fe-N interaction potential was fitted to quantum-mechanical ligand-heme energy profiles. We docked compounds into prepared CYP1A2/2HI4, CYP2C9/5X24, CYP2D6/4XRZ and CYP3A4/4K9W structures. The inputs comprise 88 global docking values, 564 named residue-energy terms, 8,126 expanded interaction terms and 1,528 pose-geometry values.

These inputs describe van der Waals interactions, electrostatics, hydrogen bonding, solvation, strain, residue contacts, aromatic geometry and heme-iron distances. Their use in ML-augmented CYP docking follows Weiser et al., 2023. Each endpoint model can use all four docking isoforms. We matched poses and records by compound identity and energy agreement, and retained compounds with incomplete coverage.

Connecting potency predictions to compound design

The components address three practical decisions. Canopy provides an inhibition-likelihood score at the defined 10 μM threshold. The regression model estimates pIC50 for each compound across all four CYPs. FITTED provides docked orientations and interaction energies that chemists can inspect alongside those predictions.

For an analogue series, a chemist can compare predicted potency and inspect whether moving a basic nitrogen changes acidic-residue contacts, whether a substituent changes aromatic packing, or whether added bulk creates a clash. Comparing poses across isoforms helps identify modifications intended to reduce unwanted inhibition or alter selectivity. This structural information gives chemists specific changes to evaluate experimentally, alongside a numerical potency estimate (Weiser et al., 2023).

Results and interpretation of ST-RAE

The OpenADMET scoring implementation defines ST-RAE as error outside experimental tolerance intervals, normalized by the same error of a constant predictor at the endpoint's mean measured pIC50. A score below 1 improves on that reference; a score of 0 means predictions fall within the experimental intervals.

Our macro ST-RAE of **0.5455** corresponds to **45.45% lower normalized interval error, averaged across the four isoforms**, than the endpoint-mean reference predictors. Each isoform scored below 1. CYP2C9 and CYP3A4 achieved ST-RAEs of 0.3817 and 0.4067, respectively. These results quantify performance against a defined reference and support compound prioritization alongside the structural analysis. The table reports the verified leaderboard snapshot retrieved on 2 November 2026.

Endpoint	ST-RAE	MAE	R-squared	Spearman rho	Kendall tau-b
CYP1A2	0.6614	0.9075	0.3664	0.7912	0.5867
CYP2C9	0.3817	0.4426	0.6583	0.8367	0.6572
CYP2D6	0.7321	1.3028	-0.0585	0.4957	0.3444
CYP3A4	0.4067	0.4745	0.7291	0.8487	0.6690
Overall	0.5455	0.7818	0.4238	0.7431	0.5643

Key publications

Labarre A. et al. *Docking Ligands into Flexible and Solvated Macromolecules. 8. Forming New Bonds-Challenges and Opportunities.* **JCIM** 2022, 62, 1061-1077. DOI.

Weiser B. et al. *Machine learning-augmented docking. 1. CYP inhibition prediction.* **Digital Discovery** 2023, 2, 1841-1849. DOI.

Forthcoming article

We are preparing a longer article on this challenge, covering model development, validation and isoform-specific chemical and residue features. We also plan to integrate Canopy inhibition scoring, pIC50 regression and FITTED pose analysis into a unified compound-assessment workflow for medicinal chemistry.